Topical Event Macroeconomic Assessment (TEMA)

An NLP-Based Framework for Mapping News Shocks to India's Sectoral Outlook

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Abstract—The Topical Event Macroeconomic Assessment (TEMA) framework is an NLP-based system designed to address critical gaps in India's macro-economic and trade intelligence. It automatically fetches news from over 50 RSS feeds daily, applies multi-layer filtering (keyword pre-filter followed by LLM relevance verification and semantic deduplication), scrapes full article content when needed, classifies events using a structured taxonomy, and generates contextual insights on sectoral impact and transmission channels. The system achieves greater than 85% relevance accuracy while maintaining data privacy through local LLM inference (gemma3:4b via Ollama). It transforms unstructured news into structured, high-fidelity signals ready for downstream economic analysis, policy monitoring, and investment decision support. This paper presents the complete architecture, pipeline design, taxonomy, real-world case studies, and future roadmap for evolving TEMA into a predictive macroeconomic decision intelligence platform.

Index Terms—Natural Language Processing, Macroeconomic Intelligence, News Analysis, Event Classification, Large Language Models, India Economy, Information Extraction, Decision Support System, Web Scraping, Retrieval-Augmented Generation.

I. INTRODUCTION

Financial analysts and policymakers in India face significant challenges in processing the overwhelming volume of global news and its rapid impact on the domestic economy. Geopolitical shocks, trade policy changes, commodity price fluctuations, and central bank decisions can significantly affect India's currency (INR), inflation, capital markets, and sectoral output within hours. However, the current process of manually monitoring, filtering, and analyzing news is slow, error-prone, and lacks structured mapping to economic indicators.

Existing news aggregation tools provide raw feeds without India-specific relevance filtering, sectoral classification, or impact assessment. This creates a critical gap between the firehose of global information and actionable macroeconomic intelligence.

This paper presents TEMA (Topical Event Macroeconomic Assessment), a production-oriented NLP-based framework that automatically collects, filters, classifies, and analyzes news to generate structured, India-focused economic insights. TEMA uses a multi-layer pipeline combining curated keywords, local Large Language Models (LLMs), and semantic processing to achieve high accuracy while preserving data privacy.

The main contributions of this work are:

- A robust multi-stage pipeline for news ingestion, cleaning, and relevance verification.
- A comprehensive hierarchical taxonomy linking news events to macroeconomic indicators.
- Real-world case studies demonstrating high, medium, and low impact event classification.
- A privacy-first architecture using local LLM inference (Ollama + gemma3:4b).
- A clear roadmap for evolving from reactive news monitoring to predictive decision intelligence using RAG and historical memory.

II. PROBLEM STATEMENT

Financial analysts and policymakers face four fundamental challenges in India's macro-economic and trade intelligence landscape:

- **Global-to-Local Volatility Propagation:** Geopolitical shocks and global macroeconomic changes (interest rate shifts, oil prices, trade wars) rapidly propagate to India's INR, inflation (CPI/WPI), Current Account Deficit (CAD), FII flows, and equity markets (Nifty/Sensex).
- **Information Overload and Redundancy:** Analysts are overwhelmed by massive volumes of repetitive and low-value news, making it extremely difficult to extract high-fidelity market signals in a timely manner.
- **Lack of Structured Taxonomy and Impact Metrics:** Most news arrives unstructured, without clear categorization by event type, sector, geographic origin, or India-specific economic impact estimates.
- **Insight Latency and Manual Overhead:** The manual process of finding, cleaning, summarizing, and validating news introduces critical delays, preventing timely reactions in high-frequency economic environments.

III. TEMA FRAMEWORK AND PIPELINE

TEMA processes raw news through a carefully engineered multi-stage pipeline designed for reliability, accuracy, and scalability:

- **Raw Input:** News enters via RSS feeds (50+ sources including Bloomberg, Reuters, BBC, WION), web search, or manual uploads.
- **Keyword Pre-Filtering:** Headlines are screened against a curated dictionary of economic and trade-related keywords.
- **LLM Relevance Verification:** A local LLM (gemma3:4b) performs contextual analysis to confirm India-specific economic relevance.
- **Semantic Deduplication:** Repeated or highly similar stories are identified and removed using semantic similarity.
- **Full Content Scraping:** Robust fallback mechanisms (requests + BeautifulSoup) with automatic retry, backoff, and politeness delays fetch full article text.
- **Event Classification:** The LLM assigns L1 Event Class and L2 Sub-type according to a comprehensive taxonomy.
- **Sentiment and Channel Mapping:** Direction (Positive/Negative/Neutral) and transmission Channel are determined.
- **Geographic Tagging:** spaCy NER extracts locations, which are mapped to countries using geomanes.
- **Database Deduplication:** Jaccard similarity check ensures only truly new content is stored.
- **Persistence:** High-quality articles with full metadata are saved for downstream analysis and reporting.

This pipeline achieves greater than 85% relevance accuracy while maintaining full data privacy through local LLM inference.

IV. KEYWORD AND LLM-DRIVEN ANALYSIS

Keywords serve as the first layer of intelligence in TEMA, helping the LLM understand the topic quickly before analyzing the full article content for accurate output. The process works as follows:

- **Keyword Matching:** The system scans news titles and summaries for predefined keywords (e.g., Oil, Fed, Tariff, RBI, Iran, Geopolitics).
- **Relevance Scoring:** Keywords help decide if the news is relevant to India (is_relevant = 1) or not relevant (is_relevant = 0).
- **Theme Identification:** Keywords help the LLM quickly understand the broad category such as Oil Shock, Fed Policy, Trade Policy, or Geopolitics.
- **Event Classification:** Keywords assist in suggesting the event_class and sub_type (e.g., Commodity_Shock, Tariff_change).

- **Initial Context for LLM:** Keywords give the LLM a quick head start to understand the topic before reading the full scraped article.
- **Full Content Analysis:** After keyword matching, the LLM reads the complete scraped article to generate accurate Insight and Channel mapping.

Key Learning: Keywords are good for initial filtering and fast triage, but full article content is mandatory for generating accurate and context-aware insights and impact assessments.

V. CORE TECHNOLOGY STACK

All components are designed for production reliability, data privacy, and scalability.

Backend & Data Layer:

Django (Python) with Django ORM and SQLite for secure authentication, custom API endpoints, and background scraper tasks.

AI & NLP Layer:

Local LLM via Ollama (gemma3:4b) for privacy-first inference at zero API cost. Performs sentiment analysis, taxonomy classification, and geographic mapping. spaCy (en_core_web_sm) handles Named Entity Recognition for location extraction.

Data Pipeline:

feedparser for RSS ingestion, requests + BeautifulSoup for robust scraping, with dynamic politeness delays (8–12s) and auto-retry logic on HTTP errors (403/429/502).

VI. WHY OLLAMA FOR LOCAL INFERENCE

TEMA uses Ollama with the gemma3:4b model for all LLM inference. This choice provides several critical advantages for a production system:

- **Zero API Cost —** No recurring token charges, enabling cost-effective scaling and experimentation.
- **Data Privacy —** All AI processing runs locally on the server; sensitive news data never leaves the system.
- **No Rate Limits —** Faster batch processing of large volumes of news without API throttling.
- **Efficient Models —** Supports lightweight yet capable models like Gemma 3:4B for quick and accurate inference.
- **Offline Capability —** AI analysis continues even without internet access.
- **Easy Customization —** Model files allow tailored system prompts and specialized workflows for macroeconomic analysis.

Result: A low-cost, private, scalable, and reliable AI backend perfectly suited for continuous news intelligence processing.

VII. EVENT TAXONOMY AND CLASSIFICATION

TEMA employs a hierarchical taxonomy to categorize news shocks and link them directly to macroeconomic indicators:

Event Classes (L1):

Trade Policy • Geo-political • Commodity Shock • Financial Market • Climate/Natural/Environment • Domestic Policy

Sub-types (L2 examples):

Tariff Change, Non-Tariff Barrier, Sanctions, Armed Conflict, Crude Oil Spike, Fed Rate Hike, Poor Monsoon, Fiscal Stimulus, RBI Rate Change, etc.

Macro Indicators Tracked:

CAD, Export Growth, INR/USD, Crude Oil Price, FII Flows, CPI Inflation, GDP Growth, Fiscal Deficit, Food CPI, Agri Output, Sectoral Output, and others.

Each classified event is enriched with Sector, Channel (transmission path), Direction, and a detailed Insight narrative explaining India-specific economic implications.

VIII. CASE STUDIES: HIGH, MEDIUM, AND LOW IMPACT EVENTS

TEMA was evaluated on real-world news events. Three representative cases are presented below.

A. High Impact Event — Commodity Shock

Field	Value
Date	07 Jun 2026
Source	Bloomberg Markets
Headline	Oil Jumps as Iran and Israel Exchange Fire With Truce at Risk
Event Class	Commodity Shock
Sub-type	Geopolitical Oil Supply Shock (Middle East Conflict, Oil Prices)
City/Country	Iran
Direction	Negative
Sector	Oil Gas and Refining
Channel	Geopolitical Oil Supply Shock & Commodity Prices & Imported Inflation & Current Account Deficit & INR Pressure

Insight:

The exchange of fire between Iran and Israel, coupled with stalled ceasefire talks, has triggered a surge in oil prices. This disruption in the Middle East raises concerns about the stability of global energy supplies. India's economy is highly susceptible to global oil price shocks. Rising oil prices directly increase India's import costs, worsening its trade deficit and fueling inflation. This will force the RBI to likely raise interest rates to combat inflation, which will slow economic growth. The corporate sector, particularly energy-intensive industries, will face reduced profitability. The geopolitical uncertainty will also negatively impact investor sentiment and financial market stability, making this a decidedly negative development for India's economic outlook.

B. Medium Impact Event — Trade Policy / Defence

Field	Value
Date	22 Jun 2026
Source	Hindustan Times
Headline	US notifies $428 million sale of support services for Apache helicopters, M777A2 howitzers to India
Event Class	Trade Policy
Sub-type	Trade Defence (US-India Defence Sales, Defence Framework)
City/Country	India / US
Direction	Positive
Sector	Defense and Aerospace
Channel	Trade competitiveness & Current account balance & INR exchange rate

Insight:

The United States has notified Congress of a $428 million sale of support services to India, encompassing logistics, personnel training, and repairs for existing Apache helicopters and M777A2 howitzers. This transaction aims to sustain India's defense platforms and bolster its capabilities to address current and future threats, particularly within the Indo-Pacific and South Asia regions. The sale underscores a strengthened U.S.-Indian strategic relationship and reflects ongoing efforts to deepen bilateral security ties. This sale is expected to proceed unless Congress objects, representing a significant investment in India's defense capabilities despite broader geopolitical tensions. The sale of support services will bolster India's defense capabilities, potentially reducing reliance on foreign maintenance and increasing domestic defense industry support, contributing to economic growth.

C. Low Impact Event — Global Factors

Field	Value
Date	22 Jun 2026
Source	Bloomberg Markets
Headline	Emerging-Market Stocks Mark Fresh High as US-Iran Talks Progress, Oil Falls
Event Class	Global Factors
Sub-type	Global Risk (Oil Prices, Emerging Markets)
City/Country	Iran / US
Direction	Neutral
Sector	Stock Market and Capital Markets (Bear Market, Sensex, Nifty)
Channel	Current account balance & INR exchange rate & Equity market

Insight:

Emerging-market stocks reached record highs alongside a significant drop in crude oil prices, fueled by encouraging progress in US-Iran peace talks following recent threats of military action. The MSCI developing markets index surged as much as 1.3%, driven by technology shares and AI optimism. While lower oil prices could benefit India through reduced import costs and potentially dampen inflationary pressures, the global shift in investor focus towards AI and the continued weakness in Chinese equities presents challenges for India's export-oriented growth strategy.

IX. IMPROVING ACCURACY WITH RAG + LARGER DATABASE

Current challenges with keyword-heavy logic include frequent hallucinated Insights, wrong Channel assignments, limited historical context, and inconsistent accuracy.

By implementing Retrieval-Augmented Generation (RAG) with a larger historical database, the system will:

- Retrieve relevant past news articles and their correct Channels/Insights from the database when new news arrives.
- Augment the LLM prompt with this retrieved historical context.
- Generate context-aware Insights and Channels grounded in real historical patterns rather than just keywords or internal model knowledge.
- Significantly reduce hallucinations and improve Channel classification accuracy.

RAG Workflow: New News Article → Retrieve Similar Past News + Correct Channels & Insights → LLM Generates Accurate Insight & Channel using Retrieved Context

X. FUTURE ROADMAP

TEMA is positioned to evolve from a prototype news aggregator into an enterprise-grade real-time macroeconomic intelligence platform. The following enhancements are planned:

1. Getting Better News Data

Current free RSS feeds are slow and often miss important details or metadata. In the future, TEMA will integrate professional paid news services such as Bloomberg, Reuters Terminal, Event Registry, and LexisNexis. These sources will provide lower latency, full-text content, historical depth, and multi-language coverage.

2. Using Much Smarter AI Models

While the current small local model (gemma3:4b) is fast, it can make mistakes on complex or nuanced news. Future versions will upgrade to larger, more capable models such as Llama 70B, Claude 3.5, or DeepSeek-R1 (671B MoE with native Chain-of-Thought). A smart routing strategy will be implemented: a small fast model will first triage incoming news, and only high-importance items will be sent to the larger model for deeper analysis.

3. Teaching AI to Think Better

Chain-of-Thought (CoT) prompting will be introduced so the model outputs a reasoning block first, followed by the final structured fields. Few-shot examples combined with structured output enforcement (using Pydantic/Instructor) will ensure consistent JSON output covering fields such as relevance_score, sector, sentiment, impact score (-5 to +5), policy_implication, and affected_indicators.

4. Making the System Smarter with Memory

The system will be given historical memory. It will remember past news and compare new events with old ones (e.g., "This rate hike is bigger than the one in December 2025"). Retrieval-Augmented Generation (RAG) with a vector database (using embeddings such as text-embedding-3-large in pgvector, Pinecone, or Qdrant) will enable comparative analysis across historical events, dramatically improving context awareness and reducing hallucinations.

Strategic Goal: Transform reactive news monitoring into predictive, context-aware macroeconomic decision intelligence.

XI. CONCLUSION

The TEMA framework successfully demonstrates a practical, privacy-preserving, and production-ready approach to converting unstructured global news into structured, India-specific macroeconomic intelligence. By combining curated keywords with local LLM reasoning, robust scraping, and a well-defined event taxonomy, the system delivers timely, high-accuracy signals that directly address the challenges of volatility propagation, information overload, and insight latency faced by analysts and policymakers.

With planned enhancements in data sources, model scale, Chain-of-Thought reasoning, and Retrieval-Augmented Generation over historical events, TEMA is positioned to evolve from a reactive news aggregator into a predictive decision-support platform capable of anticipating sectoral impacts and supporting proactive economic policy and investment strategies.

XII. REFERENCES

[1] R. Mitchell, Web Scraping with Python: Collecting Data from the Modern Web, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2018.

[2] Beautiful Soup Documentation, "Beautiful Soup 4," https://www.crummy.com/software/BeautifulSoup/, Accessed: June 2026.

[3] feedparser Documentation, "Universal Feed Parser for Python," https://feedparser.readthedocs.io/en/latest/, Accessed: June 2026.

[4] Requests Documentation, "Requests: HTTP for Humans," https://requests.readthedocs.io/en/latest/, Accessed: June 2026.

[5] NewsAPI, "News API – JSON API for live news and blog articles," https://newsapi.org, Accessed: June 2026.

[6] Ollama, "Ollama: Run large language models locally," https://ollama.com, Accessed: June 2026.

[7] spaCy, "Industrial-strength Natural Language Processing in Python," https://spacy.io, Accessed: June 2026.

[8] Django Software Foundation, "Django Web Framework Documentation," https://docs.djangoproject.com, Accessed: June 2026.